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NCE Exam Practice

PREDICTED SPECIMEN PAPER

2026

BIOLOGY

Grade 9 NCE | Paper 4

Time Allowed: 45 minutes **Total Marks:** 50

Topics covered: Disease & Immunity • The Nervous System

Candidate Name: _____

Index Number: _____

Class / Form: _____

Instructions to Candidates:

1. Write your name, index number and class in the spaces provided above.
2. Answer **ALL** questions in **ALL** sections.
3. For Section A, tick the box next to the letter of the best answer.
4. Write your answers for Sections B and C in the answer spaces provided.
5. Show all working where required — method marks may be awarded.
6. Do not use correction fluid; cross out errors neatly and rewrite.
7. The mark scheme is included at the end for self-assessment after attempting the paper.

Section A: 15 marks | **Section B:** 20 marks | **Section C:** 15 marks | **Total:** 50 marks

SECTION A — Multiple Choice Questions (15 marks — 1 mark each)

For each question, tick the box next to the letter of the **ONE** best answer.

1. Which type of pathogen causes diseases such as influenza and COVID-19? [1]
- ☐ A) Bacteria
- ☐ B) Fungi
- ☐ C) Virus
- ☐ D) Protozoan
2. The body's first line of defence against pathogens includes: [1]
- ☐ A) Antibodies
- ☐ B) Phagocytes
- ☐ C) Skin and mucous membranes
- ☐ D) T-lymphocytes
3. Antibodies are produced by: [1]
- ☐ A) Red blood cells
- ☐ B) Platelets
- ☐ C) B-lymphocytes
- ☐ D) Phagocytes
4. Which of the following best describes active immunity? [1]
- ☐ A) Receiving ready-made antibodies from another organism
- ☐ B) The body producing its own antibodies in response to an antigen
- ☐ C) Immunity passed from mother to baby
- ☐ D) Immunity from taking antibiotics
5. Vaccines work by: [1]
- ☐ A) Killing all pathogens in the body
- ☐ B) Introducing a harmless antigen to stimulate antibody production
- ☐ C) Providing antibiotics to fight infection
- ☐ D) Replacing damaged white blood cells
6. Antibiotics are effective against: [1]
- ☐ A) Viruses only

- ☐ B) Both bacteria and viruses
- ☐ C) Bacteria only
- ☐ D) Fungi only

7. Which part of the neuron carries impulses towards the cell body?

[1]

- ☐ A) Axon
- ☐ B) Myelin sheath
- ☐ C) Dendrite
- ☐ D) Synapse

8. The junction between two neurons is called a:

[1]

- ☐ A) Dendrite
- ☐ B) Synapse
- ☐ C) Axon terminal
- ☐ D) Node of Ranvier

9. A reflex action is:

[1]

- ☐ A) A slow, voluntary response to a stimulus
- ☐ B) A rapid, automatic response to a stimulus
- ☐ C) A response initiated by the brain
- ☐ D) A learned behaviour

10. The correct sequence for a reflex arc is:

[1]

- ☐ A) Effector → Motor neuron → Spinal cord → Sensory neuron → Receptor
- ☐ B) Receptor → Sensory neuron → Spinal cord → Motor neuron → Effector
- ☐ C) Receptor → Motor neuron → Spinal cord → Sensory neuron → Effector
- ☐ D) Effector → Sensory neuron → Spinal cord → Motor neuron → Receptor

11. Which part of the brain controls balance and coordination?

[1]

- ☐ A) Cerebrum
- ☐ B) Medulla oblongata
- ☐ C) Cerebellum
- ☐ D) Hypothalamus

12. The myelin sheath around a neuron functions to:

[1]

- ☐ A) Produce neurotransmitters

- ☐ B) Speed up transmission of nerve impulses
- ☐ C) Connect neurons at the synapse
- ☐ D) Produce hormones

13. Transmission of a nerve impulse across a synapse occurs via:

[1]

- ☐ A) Electrical current
- ☐ B) Direct cell-to-cell contact
- ☐ C) Chemical neurotransmitters
- ☐ D) Blood plasma

14. Which of the following is a vector-borne disease?

[1]

- ☐ A) Tuberculosis
- ☐ B) Cholera
- ☐ C) Malaria
- ☐ D) Influenza

15. Natural passive immunity is acquired by:

[1]

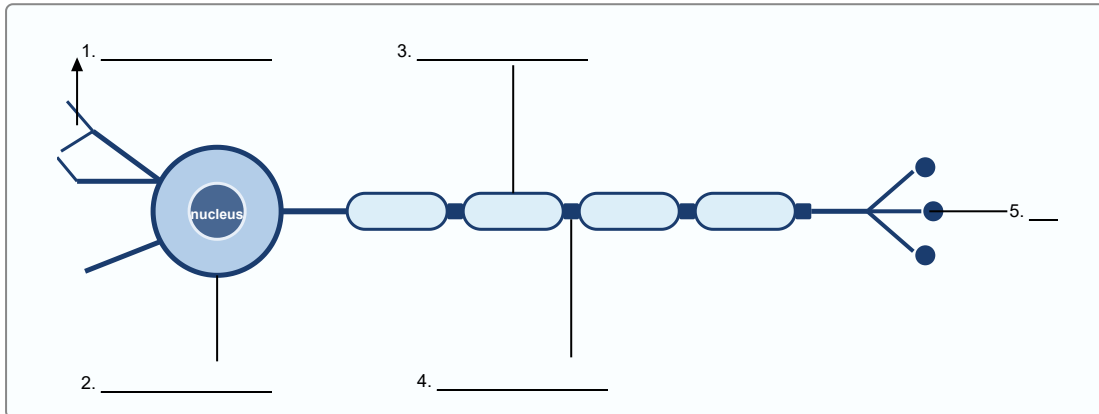
- ☐ A) Vaccination
- ☐ B) Having a disease
- ☐ C) Receiving antibodies from mother through breast milk or placenta
- ☐ D) Exposure to antigens

SECTION B — Structured Questions (20 marks — 4 questions × 5 marks each)

Answer **ALL** questions. Write your answers in the spaces provided.

B1 — Label the Neuron Diagram [5 marks]

The diagram below shows a motor neuron. Label the five structures indicated by the arrows.



Labels to use: cell body (soma) • dendrites • myelin sheath • node of Ranvier • axon terminals [1 mark each]

B2 — The Immune Response [5 marks]

(a) Distinguish between a *pathogen* and an *antigen*.

[2]

(b) Describe how phagocytes destroy pathogens.

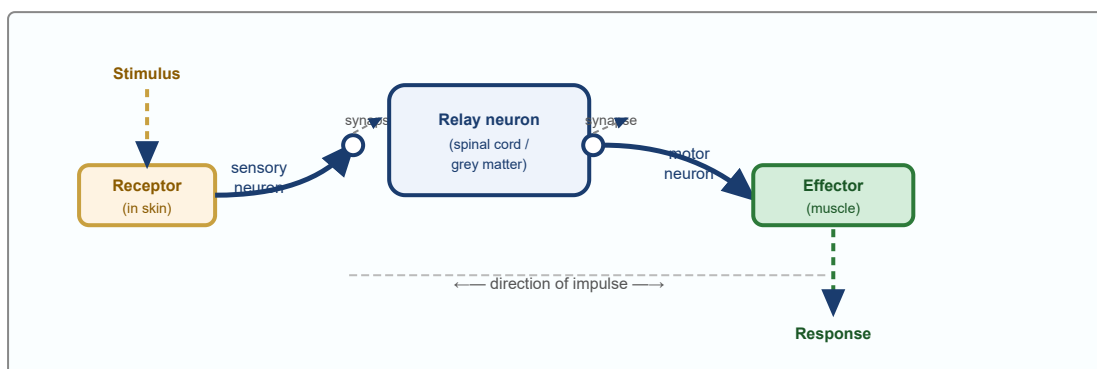
[2]

(c) Why is a second infection by the same pathogen usually less severe?

[1]

B3 — The Reflex Arc [5 marks]

Study the diagram of a simple reflex arc below.



- (a) Name the type of neuron that connects the sensory neuron to the motor neuron in the spinal cord. [1]

- (b) Explain why a reflex action is faster than a voluntary action. [2]

- (c) Give ONE example of a reflex action and explain its survival value. [2]

B4 — Disease: Types and Prevention [5 marks]

- (a) Classify each of the following diseases as *bacterial*, *viral*, or *caused by a protozoan*. [3]

Disease	Type of pathogen
(i) Malaria	

(ii) Tuberculosis (TB)	
(iii) Influenza	

(b) State **TWO** ways in which the spread of cholera can be prevented.

[2]

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SECTION C — Extended Questions (15 marks)

Answer **ALL** questions. Write in full sentences where appropriate.

C1 — The Nervous System [8 marks]

- (a) Describe the structure and function of the **THREE** types of neurons involved in a reflex arc. [3]

- (b) Explain how a nerve impulse is transmitted across a synapse. [3]

- (c) State ONE difference between the central nervous system and the peripheral nervous system. [1]

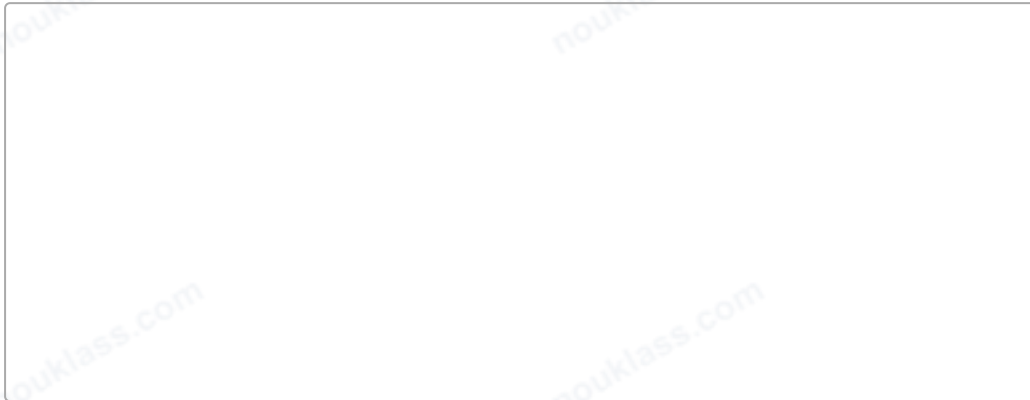
- (d) Name TWO parts of the brain and state one function of each. [1]

Part of brain	Function

C2 — Disease and Immunity [7 marks]

- (a) Explain the difference between active and passive immunity. Give ONE example of each.

[3]



- (b) Describe how a vaccine provides protection against a disease. Include reference to memory cells.

[2]



- (c) Antibiotic resistance is a growing global problem. Explain how antibiotic resistance develops in bacteria.

[2]



MARK SCHEME — Paper 4 (Biology)

For teacher/tutor use only. Accept any equivalent correct biological answer. Points marked with 'OR' are acceptable alternatives. Do not deduct marks for minor spelling errors if the meaning is clear.

SECTION A — Multiple Choice Answers

Q	Answer	Q	Answer	Q	Answer
1	C — Virus	2	C — Skin and mucous membranes	3	C — B-lymphocytes
4	B — Body producing its own antibodies	5	B — Harmless antigen to stimulate antibody production	6	C — Bacteria only
7	C — Dendrite	8	B — Synapse	9	B — Rapid, automatic response to a stimulus
10	B — Receptor → Sensory neuron → Spinal cord → Motor neuron → Effector	11	C — Cerebellum	12	B — Speed up transmission of nerve impulses
13	C — Chemical neurotransmitters	14	C — Malaria	15	C — Receiving antibodies from mother through breast milk or placenta

SECTION B — Structured Questions

B1 — Neuron Diagram Labels		
Label	Accepted Answer	Marks
1	Dendrites (OR dendrite)	1
2	Cell body (OR soma / cell body (soma))	1
3	Myelin sheath	1
4	Node of Ranvier	1
5	Axon terminals (OR synaptic knobs / terminal boutons)	1

B2 — The Immune Response		
Part	Accepted Answer	Marks
(a)	A pathogen is a microorganism (e.g. bacterium, virus) that causes disease (1). An antigen is a specific molecule (usually	2

	a protein) found on the surface of a pathogen that is recognised as foreign by the immune system and triggers an immune response (1).	
(b)	Phagocytes detect and move towards pathogens by chemotaxis (1). They engulf the pathogen by phagocytosis (changing shape to surround it), then enzymes within lysosomes are released to digest and destroy the pathogen (1).	2
(c)	Memory cells (B-lymphocytes) produced during the first infection remain in the blood (1). On second infection, these memory cells recognise the same antigen and produce antibodies much more rapidly and in larger quantities, clearing the infection before serious symptoms develop.	1

B3 — The Reflex Arc

Part	Accepted Answer	Marks
(a)	Relay neuron (OR interneuron / connector neuron).	1
(b)	In a reflex action the nerve impulse passes through the spinal cord only — it does not need to travel all the way to the brain for processing (1). This shorter pathway reduces the time taken for the response; voluntary actions require signals to be sent to and processed by the brain before a response is sent back, which takes considerably longer (1).	2
(c)	Example: withdrawing your hand from a hot object (pain withdrawal reflex). Survival value: it removes the body part from danger before the brain has time to consciously process the stimulus, minimising tissue damage (1 mark for example; 1 mark for survival value).	2

B4 — Disease: Types and Prevention

Part	Accepted Answer	Marks
(a) (i)	Malaria — caused by a protozoan (<i>Plasmodium</i>).	1
(a) (ii)	Tuberculosis (TB) — caused by bacteria (<i>Mycobacterium tuberculosis</i>).	1
(a) (iii)	Influenza — caused by a virus (influenza virus).	1
(b)	Any TWO of: (1) Provide clean, treated drinking water / boil water before drinking. (2) Improve sanitation — proper sewage disposal to prevent contamination of water supplies. (3) Wash hands thoroughly with soap after using the toilet and before handling food. (4) Cook food properly and avoid consuming contaminated food. (5) Oral rehydration therapy for infected individuals to prevent further spread via poor sanitation.	2

SECTION C — Extended Questions

C1 — The Nervous System [8 marks]

Part	Accepted Answer	Marks
(a)	Sensory neuron: carries impulses FROM a receptor (e.g. skin) TOWARDS the spinal cord / CNS; has a long dendron and short axon (1).	3

	Relay neuron (interneuron): found within the spinal cord / brain; connects sensory neurons to motor neurons; processes and relays the impulse (1). Motor neuron: carries impulses FROM the spinal cord / CNS TO an effector (muscle or gland); has a long myelinated axon (1).	
(b)	When an impulse arrives at the pre-synaptic terminal (axon terminal), it triggers vesicles containing neurotransmitters to fuse with the pre-synaptic membrane (1). Neurotransmitters are released by exocytosis into the synaptic cleft / gap (1). The neurotransmitters diffuse across the cleft and bind to specific receptor sites on the post-synaptic membrane, generating a new electrical impulse in the next neuron (1).	3
(c)	The central nervous system (CNS) consists of the brain and spinal cord; the peripheral nervous system (PNS) consists of all the nerves connecting the CNS to the rest of the body. (Accept any one clear structural OR functional difference.)	1
(d)	Any TWO correct pairs (award 0.5 per correct pair, but as the question is [1 mark], award 1 if at least two correct): Cerebrum — conscious thought, memory, language, voluntary movement / Cerebellum — balance and coordination / Medulla oblongata — controls heart rate, breathing rate, blood pressure / Hypothalamus — regulates body temperature, hunger, thirst.	1

C2 — Disease and Immunity [7 marks]		
Part	Accepted Answer	Marks
(a)	Active immunity: the body's own immune system produces antibodies in response to exposure to an antigen; it is long-lasting because memory cells are produced (1). Example: recovering from an infection (natural active immunity) OR being vaccinated (artificial active immunity) (1). Passive immunity: ready-made antibodies are introduced into the body; it is short-lived because no memory cells are produced (1). Example: antibodies passed from mother to baby via placenta or breast milk (natural passive) OR injection of antitoxin / antiserum (artificial passive) (accept one example).	3
(b)	A vaccine contains a harmless form of a pathogen (dead/weakened pathogen, antigen, or mRNA coding for an antigen) which is introduced into the body (1). The immune system responds by producing antibodies and memory B-cells specific to that antigen. If the real pathogen later infects the body, the memory cells rapidly multiply and produce large amounts of antibodies before the pathogen can cause serious illness (1).	2
(c)	Within a population of bacteria there is natural genetic variation; some bacteria may have a random mutation that makes them resistant to an antibiotic (1). When antibiotics are used, non-resistant bacteria are killed while resistant ones survive and reproduce (natural selection), passing on the resistance gene to offspring — so the proportion of resistant bacteria in the population increases over time (1).	2

Mark Summary: Section A: 15 | Section B: 20 | Section C: 15 | **Total: 50 marks**

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